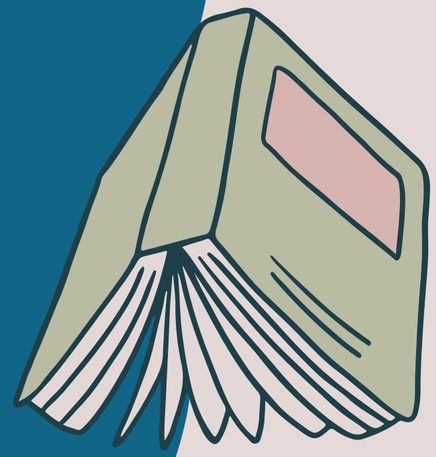
ONLINE BOOK STORE MANAGEMENT



TEAM MEMBERS

1.Parimala.N
2.Manvitha rai.R

PROBLEM STATEMENT

Managing bookstore data becomes difficult when

1. Many categories exist
2. Books belong to different categories
3. Hierarchical structure is needed

Real-World Applications

1. Amazon / Flipkart category system

2. Library management systems

3. File system (folders & files)

Data Structure Used

Tree (General Tree)

1. Child pointer → first child
2. Sibling pointer → next node

SYSTEM DESIGN

1. Root → Online Bookstore

2. Categories → Child nodes

3. Books → Leaf nodes

Operations (CRUD)

1. Create → Add Node
2. Read → Display Tree
3. Update → Modify Name
4. Delete → Remove Node
5. Search → Find Node

Algorithm

1. Start
2. Display menu
3. Perform operation
4. Repeat until exit

Implementation

1. Language: C
2. Dynamic memory allocation (malloc)
3. Recursive traversal (DFS)

Demo Output

=> [0] Online Bookstore

=> [1] Fiction

=> [2] Harry Potter

Conclusion

1. Tree structure simplifies hierarchy

2. Scalable for large systems

3. The system efficiently organizes books and categories using a TREE data structure, making data easy to manage